# REDUCEDNESS OF A NOETHERIAN LOCAL RING AND OF ITS COMPLETION

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Allowed standard prerequisites (editorial).*

- Reducedness descends along faithfully flat ring maps, and the completion map is faithfully flat. Algebra, Lemma \ref{algebra-lemma-descent-reduced}, cited at proof lines 10710-10711. Gives the first assertion.
- An example of a reduced local ring whose completion is not reduced. Algebra, Example \ref{algebra-example-bad-dvr-char-p} and Algebra, Remark \ref{algebra-remark-resolution-dim-1}, cited at proof lines 10712-10714. Gives the second assertion.
- Analytic unramifiedness of local domains and the criterion for reducedness of Nagata rings. Algebra, Lemma \ref{algebra-lemma-local-nagata-domain-analytically-unramified} and Algebra, Lemma \ref{algebra-lemma-analytically-unramified-easy}, cited at proof lines 10715-10718. Give the third assertion.

*Source reference.* Original lemma 07NZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9779–9779. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 033F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4054–4054. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 00PB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 925–925. Navigation pointer to a statement the

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*Source reference.* Original lemma 00PC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 926–926. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 0331: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4040–4040. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 032Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4037–4037. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $A$ be a Noetherian local ring.*

(1) *If $A^\wedge$ is reduced, then so is $A$.*
(2) *In general $A$ reduced does not imply $A^\wedge$ is reduced.*
(3) *If $A$ is Nagata, then $A$ is reduced if and only if $A^\wedge$ is reduced.*

*Proof.* As $A \to A^\wedge$ is faithfully flat we have (1) by Algebra, Lemma 033F. For (2) see Algebra, Example 00PB (there are also examples in characteristic zero, see Algebra, Remark 00PC). For (3) see Algebra, Lemmas 0331 and 032Y. □